

DATA VORTEX (AARUUSH '26) — TECHNICAL RECOVERY & FORENSIC EDA REPORT

Phase 1: Telemetry Stream Recovery, Data Audit & Exploratory Analysis

Prepared For: Technical Evaluation Committee, Aaruush '26

System Component: Social Engine Microservices Pipeline (node_07)

1. System Outage & Architectural Context

Following an unannounced database microservice failure (**service exit code 137**, memory limit breach), the live social telemetry ingest pipeline experienced severe payload corruption across primary cluster nodes. Raw memory frame extraction from recovery node **node_07** yielded 1,500 raw user dimensional records and 12,360 post event frames.

Through our vector-accelerated ETL pipeline ([data_cleaning_pipeline.py](#)), system structural integrity was completely restored. The processing pipeline sanitized all anomalous records, producing **1,500 clean user entity records** and **12,000 deduplicated post event frames** ready for Phase 2 database schema ingestion.

2. Data Quality Diagnostics & Anomaly Resolution Strategy

During the forensic audit of the corrupted buffer dumps, four major telemetry failure modes were identified and systematically resolved:

- **Packet Duplication:** Removed 360 redundant post event frames caused by network API retry loops during system failure.
- **Heterogeneous Datetime Formats:** Standardized multi-format strings (Unix epoch integers, ISO 8601 strings, and European standard date strings) into a uniform **YYYY-MM-DD HH:MM:SS** standard.
- **Sign-Bit Telemetry Errors:** Reversed 509 negative engagement metric records (likes, dislikes, shares, etc.) caused by integer sign-bit flipping during memory page dumps using absolute value transformations.
- **Missing Value Imputation:** Imputed missing interaction values using platform-grouped medians to preserve empirical variance and prevent metric distribution skew.

3. Summary Statistics of Recovered Telemetry

Below is the verified post-recovery statistical summary across all primary interaction and entity metrics:

Metric	Mean	Std Dev	Median	Max	Skewness
Followers	24,925.03	14,237.10	24,755.50	49,944.00	0.01
Likes	2,493.69	1,324.92	2,496.00	5,000.00	0.00
Shares	1,007.17	575.07	1,018.00	2,000.00	0.00
Comments	504.35	288.68	503.00	1,000.00	0.00
Total Engagement	4,005.21	1,477.48	4,008.00	7,893.00	-0.01

4. Platform Interaction & Demographic Breakdown

A. Post Volume Distribution Across Social Channels

- **Facebook:** 2,071 posts (~17.3%)
- **YouTube:** 2,071 posts (~17.3%)
- **Twitter / X:** 2,048 posts (~17.1%)
- **Reddit:** 2,030 posts (~16.9%)
- **Instagram:** 1,988 posts (~16.6%)
- **Unknown / Unassigned:** 1,792 posts (~14.9%)

B. Channel Interactions

- Interaction density across all valid platforms displays near-uniform average engagement, averaging **~2,500 likes**, **~1,000 shares**, and **~500 comments** per post event.
- Total interaction density follows a clean normal distribution centered around **4,008 interactions per post**.

5. Architectural Recommendations for Phase 2

1. **Ingress Validation Gateways:** Implement strict schema validation at API endpoints to reject out-of-bounds negative integers and enforce ISO 8601 timestamps prior to database ingestion.

2. **Idempotency Keys:** Enforce unique transaction hashes (`post_id`) at the queue level to prevent duplicate payload insertions during network retry spikes.
3. **Primary Key Constraints:** Configure relational foreign key relationships mapping `posts.user_id` to `users.user_id` to prepare for Phase 2 SQL Schema Migration.